Improving the training and application of chinese language models for low-order grammar

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ABSTRACT

This paper presents a method to enhance the training of low-order grammar language models in response to the issue of out-of-vocabulary (OOV) words and incorrect words in acoustic model and low-order grammar language model decoding, which results in a decrease in Chinese speech recognition accuracy. To address the problem of incorrect words caused by polyphonic Chinese character, multiple phonemes are added to the Chinese dictionary to improve the phoneme model training. Prior to training the low-order grammar language model, the one-dimensional word segmentation is converted into a vectorized segmentation structure for processing. This enables the training of language models to obtain different probability distributions for the same word, thereby reducing the occurrence of OOV words. Finally, this method is compared with traditional language model training methods and evaluated with the same acoustic parameters. The experiment demonstrates an 8.5% reduction in word error rate for speech recognition when using low-order grammar language models.

Keywords: N-gram, Word Segmentation, Phoneme Model, TDNN, Speech Recognition

1. INTRODUCTION

In 2012, Hinton et al. proposed a deep neural network (DNN) acoustic model. Combining hidden Markov models (HMM) further improved recognition technology beyond traditional HMM+GMM acoustic models. In 2014, Maas et al. achieved good results by introducing maximum likelihood estimation in training a relatively simple DNN model on a 2100-hour English Switchboard benchmark corpus. In 2015, Daniel Povey et al. proposed a Time Delay Neural Network (TDNN) structure with inter-frame information based on DNN. Compared to the DNN model, the TDNN improved the recognition rate in the Switchboard task and is suitable for various task scenarios[1].

The above acoustic models, combined with language models, have achieved good results. N-gram is the most commonly used language model in speech recognition. This model takes N consecutive words in a sentence and uses maximum likelihood estimation to obtain word probabilities from the training corpus, thereby inferring sentence structure. High-order grammar has better constraint ability on sentences. Although it has better decoding performance, it affects decoding speed due to the large model size and long training time. Low-order grammar (usually N=1) has the least constraints on sentences. The word probability predicted by N-gram depends on the previous word, and the dependence on longer distance context may often be ignored due to the decrease of N, resulting in the problem of zero probability[2].

Chinese scholars have made significant achievements in various aspects of Chinese speech recognition. In 2015, a team from Tsinghua University developed THCHS30, the first open-source Chinese speech database, which includes a pronunciation dictionary (lexicon) and language models for researchers to download. Many Chinese language models are trained based on the GigaWord corpus, including the THCHS30 model.

In Chinese speech recognition, HMM+GMM, DNN+HMM, and TDNN+HMM models were used for acoustic parameter training in 2022[3-4]. The THCHS30 pronunciation dictionary and language models were introduced. The word error rates of the models showed a significant decrease compared to the former models.

From the above analysis of the recognition results, a large number of out-of-vocabulary (OOV) words and incorrect words lead to a decrease in recognition rate under N=1. Although some scholars have proposed smoothing methods[5] to solve the OOV phenomenon, the sparsity of vocabulary itself exists in low-order grammar.

Given the above background, this paper proposes two directions to improve the training method of the model: word segmentation structure and phonemes. When building the phoneme dictionary, Polyphone phonemes are added and the phoneme model is retrained to minimize the occurrence of incorrect words caused by polyphonic characters. Prior to

training the language model using low-order grammar, we first expand Chinese word segmentation into a vectorized form for processing. This allows for the possibility of expanding the word list and reducing the occurrence of out-of-vocabulary words. During the experimental process, the THCHS30 language model as the benchmark, the texts are retrained by the improved method. Finally, we compare it with the traditional training language model.

2. IMPROVING OF THE LOW-ORDER GRAMMAR MODEL

2.1 N-gram language model

Given a series of words $w_1, w_2, \ldots, w_T$, is noted $W = (w_1, w_2, \ldots, w_T)$.

According to the Markov model, $\mathrm{p}(w_1, w_2, \ldots w_T) = \prod_{i=1}^{T} \mathrm{p}(w_i \mid w_{i-1}, \ldots, w_{i-n+1})$, the N-gram language model[6] is as follows.

$$\mathrm{p}(w_i \mid w_{i-1}, \ldots, w_{i-n+1}) = \frac{C(w_{i-n+1}, \ldots, w_{i-1}, w_i)}{C(w_{i-n+1}, \ldots, w_{i-1})} \tag{1}$$

The symbol p represents the probability of any word w_i and w_i depends on its previous n words. The symbol C records the count of previous n words, including w_i. If w_i only depends on itself, it is called low-order grammar. Otherwise, it is called high-order grammar.

2.2 Improvement of N-gram

Convert W into vector form $\boldsymbol{W} = (W_1, W_2, \ldots, W_K)'$, containing K dimensions, where $W_i = (w_1, w_2, \ldots, w_T)$ is noted a series of word. The vector form similar to formula 1 is shown below.

$$\mathrm{p}(\boldsymbol{W}) = \left(\prod_{i=1}^{T} \mathrm{p}(w_i)^{C_1(w_i)} \quad \ldots \quad \prod_{i=1}^{T} \mathrm{p}(w_i)^{C_K(w_i)} \right)'$$

and $j \in \{1, 2, \ldots, K\}$

$$\mathrm{p}(w_i) = \frac{C_j(w_i)}{\sum_{i=1}^{T} C_j(w_i)} \tag{2}$$

The symbol C_j count the number of occurrences of each word in the given sequence. Formula 2 is referred to as the improvement format of the low-order grammar (N=1) to calculate the probability of any word w_i. The specific form of it will be displayed later. The following formula is called improvement format of high-order grammar ($\mathrm{N} \geq 2$).

$$\mathrm{p}(w_i \mid w_{i-1}, \ldots, w_{i-n+1}) = \frac{C_j(w_{i-n+1}, \ldots, w_{i-1}, w_i)}{C_j(w_{i-n+1}, \ldots, w_{i-1})} \tag{3}$$

2.3 Training demonstration

When K=1, the above model is precisely the traditional N-gram model. For example a Chinese sentence W =“阿拉伯数字是阿拉伯人发明的吗”(W = "Were the Arabic numerals actually invented by the Arabs ?"), count the occurrences of "the Arabic numerals", i.e. the probability p(w_i) =“阿拉伯数字”(p(w_i) = "the Arabic numerals"). Assuming the the Chinese word segmentation forms[7-8] is as follows.

K=1: <s>阿拉伯\数字\是\阿拉伯\人\发明\的\吗<\s>

(<s>Were the Arabic numerals actually invented by the Arabs ?<\s>)

K=2: <s>阿拉伯\数字\是\阿拉伯人\发明\的吗<\s>

(<s>Were the Arabic numerals actually invented by the Arabs ?<\s>)

K=3: <s>阿拉伯数字\是\阿拉伯\人\发明\的\吗<\s>

(<s>Were the Arabic numerals actually invented by the Arabs ?<\s>)

K=4: <s>阿拉伯\数字\是\阿拉伯人\发明的\吗<\s>

(<s>Were the Arabic numerals actually invented by the Arabs ?<\s>)

K=4 represents that there are 4 different Chinese segmentation structures for the same sentence *W*. Table 1 shows that increasing the likelihood of finding words in low-order grammar (N=1) can decrease the occurrence of zero probability in the training demonstration. The same result is also found in Equation 2.

Table 1. Probability of different *K* values.

	Order N			**Values** *K*
	1	2	3	
Probability p(w_i)	0	0	0.5	1
	0	0	1	2
	0.07	1	0	3
	0	0	1	4

The possibility of expand the word list increase and reduce the occurrence of out-of-vocabulary words, given the above. Figure 1 shows the improvement of the training process.

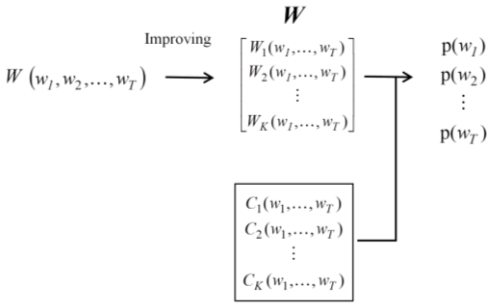

Figure 1. Improving the Training of N-gram Language Model.

3. APPLICATION OF LOW-ORDER GRAMMAR LANGUAGE MODEL

3.1 Improvement training of Chinese phonemic model

Phonemes are used as the smallest unit of speech. When constructing a phonemic dictionary, the phenomenon of incorrect words is taken into account, and polyphonic is added. Table 2 lists some Pinyin and phonemes. Polyphonic Chinese characters are easier to recognize, e.g

约 ii ao1

phoneme ii ao 1 corresponding to Pinyin yao

约 i ao1

phoneme i ao 1 corresponding to Pinyin yao

约 vv e1

phoneme vv e1 corresponding to Pinyin yue

约 ve e1

phoneme ve e1 corresponding to Pinyin yue

Table 2. Comparison of Chinese Phonemes and Pinyin.

Pinyin	Phoneme 1	Phoneme 2
zi	z	iy
yi	ii	i
sh	sh	ix
yu	vv	ve
ye	iii	ie

For the Chinese dictionary, Chinese is mapped into Pinyin, and then Pinyin is mapped into phonemes. The Chinese phoneme model is retrained by Grapheme-to-Phoneme (G2P)[9]. For the training set corpus, the pronunciation phonemes are forced to be aligned each time, and the Montreal Forced Aligner (MFA)[10]method is used to generate a pronunciation phoneme dictionary.

3.2 Language model retraining

Firstly, transform the Chinese sentence structure into a vectorized segmentation structure. Prior to the experiment, four segmentation techniques were utilized to process the text.

Ultimately, the language model parameters are trained using the SRILM tool[11]. The size of the trained model is presented in Table 3.

Table 3. Lexicon Scale.

Lexicon	N=1	N=3
3.4MB	393KB	34M

3.3 DNN architecture

Deep neural network (DNN)[12] are typically composed of three layers: the input layer, hidden layer, and output layer. The DNN architecture utilized in this paper consists of an input layer, two hidden layers, an output layer, and a classifier, totaling five layers. The Softmax classifier is employed. The DNN architecture is illustrated in Figure 2.

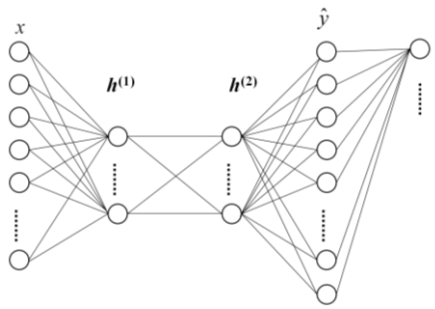

Figure 2. Deep neural network.

3.4 Experimental setting

This article selects the open-source Mandarin datasets THCHS30 and AISHELL as the training set[13-14], with a small portion of THCHS30 as the test set. To ensure the correlation of information during the training process of acoustic parameters, this paper utilizes a Delayed Neural Network with a specific structure called TDNN[15], as shown in Figure 3's TDNN module. Some training parameters are listed in Table 4, and the configuration of the TDNN network is outlined in Table 5.

Table 4. Parameter settings.

Parameter	Values
epochs	4
initial-effective-lrate	0.0015
final-effective-lrate	0.00015
minibatch	128

Table 5. Delayed Neural Network settings.

Net	Layers
TDNN[-7,5]	[-2,2]{-2,2}{-3,4}{0}{0}

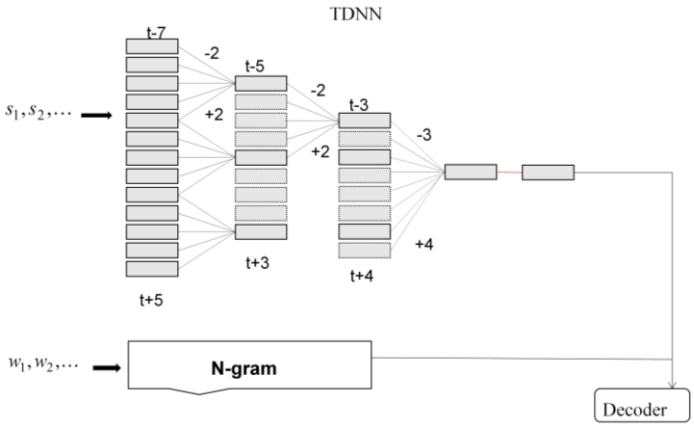

Figure 3. Decoding.

The training equipment[16] consists of an NVIDIA GeForce GTX 1080 Ti GPU, an Intel Xeon E5-2678 V3 processor, and 64GB of memory. The software environment runs on Ubuntu 15 operating system with Python 2.7.

3.5 Experimental results

Word Error Rate (WER) is a metric used to evaluate the accuracy of speech recognition. Its definition is as follows.

$$\text{WER} = \frac{E}{R} \tag{4}$$

The symbol E represents the number of incorrect words, including OOV words, and the symbol R represents the number of recognized words. When decoding at N=1 and N=3, it is important to compare the language models and the enhanced training models separately.

The experimental results indicate a decrease in the WER for each group as shown in Figure 4. Specifically, under N=1, the WER of K=4 decreased by 8.5% compared to K=1. Under N=3, the WER decreased by 1.1%. The decrease in the WER is more significant under low-order grammar, and parallel training does not significantly affect the model's training speed. Under high-order methods, the recognition rate tends to stabilize.

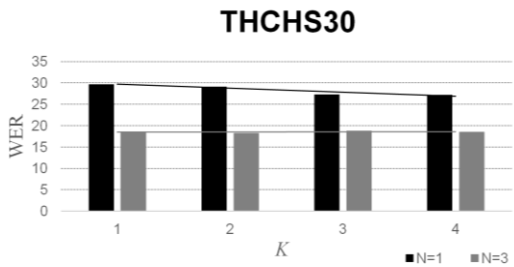

Figure 4. WER corresponding to different K when N=1 and N=3 are taken in the THCHS30 corpus.

4. CONCLUSION

Language models rely on vocabulary and phoneme dictionaries for training. This article proposes word segmentation structure and phonemes to improve the training method of the model. Try different combinations of word segmentation methods to influence the model results. Although this article do not find a reasonable probability distribution pattern, it do find that word probability distributions are not unique. According to formula 2, as the dimension of segmentation

increases, the number of words will also increase accordingly. The experimental results also confirm the above conclusion. Experimental results indicate that the WER decreases significantly under low-order grammar, while the rate is relatively stable under high-order grammar. It should be noted that this article only verifies recognition on a small-scale publicly available corpus and does not further study recognition on data sets outside the corpus. More in-depth research is needed on corpus vocabulary adaptive expansion and incorrect words.

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